

## Highlights

### **Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction**

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.541 to 0.616
- Across 20 transfer directions the thermal gain is +0.004 [-0.028, +0.036]

# Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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## Abstract

Pre-fire thermal wildfire models are rarely tested outside their fitting region. Six thermal predictors were added to a terrain, fuel and greenness baseline in five Mediterranean regions, with MCD64A1 labels and spatially blocked validation. The first result is **evaluation geometry**: with the model held fixed, scoring on the burn scar and its 2 km collar rather than region-wide costs **0.143 ROC-AUC** [**+0.077**, **+0.208**], driven by the negative pool, not class balance. Between regions, equalising the study areas to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.541 to 0.616. Local skill does not travel: the thermal block adds +0.045 to +0.148 within regions at 5 km blocking but +0.004 [−0.028, +0.036] across twenty transfer directions, and the static baseline transfers no better, at 0.537 against 0.541. Transfer skill must be measured on the target region before a model is relied on.

*Keywords:* wildfire, model transferability, model evaluation, land surface temperature, domain adaptation, concept shift

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## 1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns (Pausas and Keeley, 2021). The dominant pattern in fire susceptibility mapping (Vibhandik et al., 2026; Jodhani et al., 2026) is settled: geospatial predictors are assembled over a study region, paired with a historical burned-area record and fitted with a supervised classifier, most often a random forest (Breiman, 2001), and the surface is published with a cross-validated AUC of 0.85 to 0.95. **That pattern does not report portability**, and this paper measures it.

### 10 1.1. *Portability goes unmeasured*

A susceptibility model’s reported skill is almost always a *within-region* estimate: held-out folds come from the same study area and season, often the same fire, and random folds over cells let spatial autocorrelation inflate it further. The problem is documented across ecological modelling (Roberts et al., 2017; Ploton et al., 2020) and addressed by spatially blocked cross-validation (Valavi et al., 2019; Meyer et al., 2018), but blocking does not make the estimate honest about a fire the model has not seen.

Because only that side of the ledger is reported, portability is never entered: a predictor block is adopted on the increment it delivers inside its training footprint, and whether that survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger indices are known not to port cleanly in a Peruvian case study (Podschwit et al., 2022), and the two studies that test transfer systematically (Dimarco et al., 2026;

25 Liu and Mostafavi, 2025) both report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific class behaves the same way is this paper’s question.

### 1.2. Pre-fire thermal dryness is the natural test case

30 Pre-fire thermal dryness is the dynamic class most plausibly *expected* to transfer. Standard predictors are static or near-static over the timescale at which fire danger varies, so they explain poorly why one summer burned and the preceding one did not; what changes is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The physics  
35 linking moisture stress to combustion is universal, so portability should be most expected here, which makes the class diagnostic: a loss cannot be dismissed as a peculiarity of a locally defined covariate. The expectation is sharpest for the internally normalised channels, which should be least exposed to absolute-temperature offsets between regions; **they transfer no**  
40 **better than the absolute ones** (Appendix A(f)). The expectation motivates the design and is not a finding of it — Section 4.4 reports the associations running the other way.

### 1.3. Contributions

Two findings carry this paper, stated here as claims and established in  
45 Section 4, which carries every interval.

**Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve findings of our own.** That evaluation extent inflates AUC is established in species

distribution modelling (Section 2.4); what is new is a magnitude under a  
 50 controlled design. Holding model, predictors and fitting fixed and changing  
 only which cells are scored costs **0.143 ROC-AUC**, the size of the incre-  
 ments this design measures for a predictor block, with the cause measured as  
 the negative pool’s composition rather than class balance (Section 4.3). Ap-  
 plied between regions it withdraws five claims of our own, including the sign  
 55 reversal we had offered as the mechanism and the one arm that held place  
 fixed (Section 4.4), and the practical consequence is a reporting standard  
 (Section 5.6).

**Contribution 2. Local skill does not travel, and correcting the  
 frame does not rescue it.** The thermal block is worth a substantial within-  
 60 region increment in all five regions, every bootstrap interval above zero, and  
 stays positive when the frame is equalised — but much of it is a property of  
 interleaved holdout, and across twenty transfer directions the paired con-  
 tribution spans zero on both frames under the primary resampling unit,  
 though much closer to the boundary once equalised and not under every  
 65 admissible unit (Section 4.4), with a sign belonging to the pair rather than  
 the block. Two controls bound this: the static baseline transfers no better  
 on either frame, so the failure is not the thermal block’s peculiarity, and on  
 matched frames and blocking equalised transfer still falls **0.155** short of the  
 within-region reference. The within-region half is not novel (Alkan Akıncı  
 70 and Akıncı, 2023; Iban and Sekertekin, 2022); the paired contrast against  
 portability is.

Two consequences follow, as supporting results: label-free alignment by  
 standardisation and covariance alignment (Sun et al., 2016) does not repair

transfer, in what we believe is its first application to fire susceptibility; and  
75 because the residual is conditional, the resource that closes it is target labels,  
priced in Appendix A(u).

The leakage-audited, spatially blocked protocol is released with code,  
configuration and frozen outputs, so most of this can be re-run rather than  
taken on trust, the release being complete as the declarations record — which  
80 matters given evidence that wildfire transfer conclusions are sensitive to eval-  
uation design (Xu et al., 2026). A companion paper treats the observational  
layer beneath this one.

## 2. Related work

### *2.1. Fire susceptibility mapping, and what it reports*

85 The dominant pattern is described in Section 1; the figure it publishes,  
a cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Com-  
parable within-region results already exist for the landscapes studied here  
(Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022).

### *2.2. Pre-fire thermal dryness*

90 Satellite thermal observation gives repeated access to surface state  
through fuel moisture content (Yebra et al., 2013). Chuvieco et al. (2004)  
established the pairing of land surface temperature with a vegetation  
index as a live-fuel-moisture estimator for fire-danger rating, and the  
Temperature-Vegetation Dryness Index (Sandholt et al., 2002) formalises  
95 that feature space into an internally normalised measure, which is why  
it is often expected to travel better than raw temperature. That pre-fire

thermal state carries information about subsequent fire is established (Maffei et al., 2018; Maffei and Menenti, 2019; Maffei et al., 2021), and Gelabert et al. (2025) found dead fine fuel moisture and its anomalies the most  
100 influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether such a classifier survives strict, label-free application to an unseen region.

### *2.3. Spatial validation and transferability*

105 The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes (Moreno-Torres et al., 2012). Covariate shift is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL (Sun et al., 2016),  
110 and domain adaptation has an established remote-sensing literature (Tuia et al., 2016; Persello and Bruzzone, 2012). Concept shift is not: no realignment of inputs can repair a reversal in the sign of an association, because detecting one requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new (Huang et al., 2026), but we are aware  
115 of no prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

### *2.4. Cross-region generalisation of fire models*

Few studies test fire-model transfer directly. Podschwit et al. (2022) report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome (Liu  
120

and Mostafavi, 2025) runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. (2026) argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Appendix D states what was fixed and when. Kondylatos et al. (2023) provide Mesogeos, a 1 km Mediterranean datacube.

**Evaluation extent and AUC.** Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. (2008) make it the fifth and, by their own ranking, most important reason to distrust AUC comparatively: the extent of the modelled area governs how many easy absences enter the calculation, and therefore the score. VanDerWal et al. (2009) show the same lever on the calibration side, and Barve et al. (2011) give the argument its general form as the accessible area. That literature is qualitative about magnitude, because magnitude is problem-specific, so our Contribution 1 is a measurement inside that framework rather than a new phenomenon. The wildfire literature has largely not imported the lesson: region-wide figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells — which is why Section 5.6 treats the 0.143 as a reporting problem rather than a caveat.

**The nearest neighbour, and the contrast this paper draws.** Dimarco et al. (2026) is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full  $4 \times 4$  matrix, every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher, with no domain adaptation.<sup>1</sup> Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season.

**The response variable:** theirs is human-driven ignition, ours burned area, which have different dominant controls. Read together the two results suggest the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesik et al. (2021) and Rousseau and Betts (2022) aligning from species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement; Section 5.5 states why our data do not settle it.

### 3. Data and methods

#### 3.1. Study regions and temporal windows

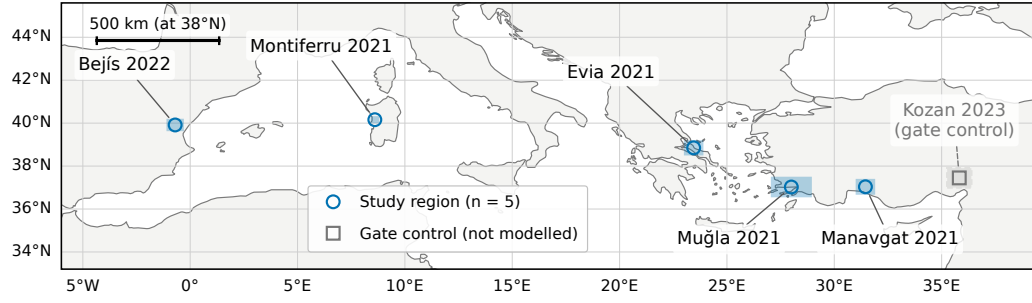
Five Mediterranean wildfire regions are analysed (Fig. 1): Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in

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<sup>1</sup>Their Results text refers to “LST anomalies”, inconsistent with their own Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology (Muñoz-Sabater et al., 2021). We follow their Methods and note the discrepancy so the comparison is transparent.

Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangle in EPSG:4326, defined from place coverage rather than from a fire perimeter and deliberately not clipped to it, so that unburned cells around each fire form the negative class. **The record does not show every box fixed before any outcome was seen**, so what it does show is stated region by region. In the pipeline’s version history each box has a single committed value, never changed afterwards. Montiferru’s is derived deterministically from the union of four municipal boundaries. Manavgat’s was drawn to exclude the coastal cropland belt and committed together with the gate-only workflow that first evaluated it, the commit that first gave the pipeline any Manavgat geometry; its registry comments disagree on whether the drawing preceded or followed that first gate run. Bejís’s is still labelled the initial candidate in the registry but was committed after its first gate and model results. Muğla’s coordinates appear in a dated preflight record about four minutes before its first gate result and were committed to the registry only afterwards. No box needed adjusting to pass the gate, whose admitted margins are wide (Section 4.1). **One choice was label-informed and is stated as such:** the North Evia box was extended after the legacy box proved atypically high in burned prevalence, the extended geometry then defined from place anchors and the legacy variant kept as a sensitivity arm. Section 4.4 shows this framing decision is consequential and Appendix C.5(ix) treats it as the design lesson of the paper. A sixth region, Kozan 2023, is carried as a negative control and excluded by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation

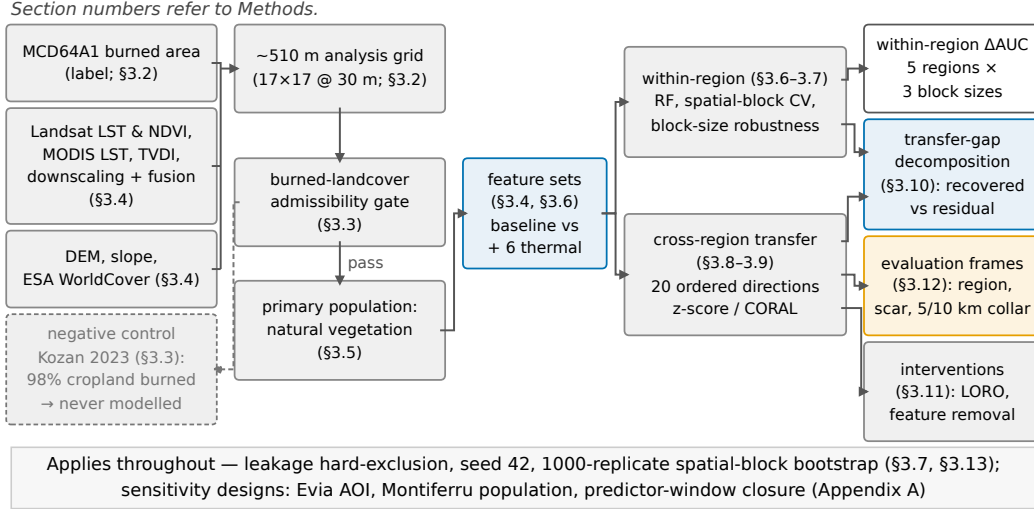


**Figure 1: Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table C1 to  $5 \times 10^{-3}$  degrees.

can post-date the first labelled burning. Lengths vary with the event: 57 to 61 days for predictors, 35 to 59 for labels, counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region  
195 identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Appendix C, Table C1. The processing chain and the evaluation programme it feeds are drawn in Fig. 2.

### *3.2. Burned-area label and the $\sim 500$ m analysis grid*

200 Labels come from MODIS MCD64A1 Collection 6.1 (Giglio et al., 2018) retrieved through Google Earth Engine (Gorelick et al., 2017), whose omission and commission characteristics (Boschetti et al., 2019) bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks,  
205 giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the label window; the label never affects eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning before the label window  
210 opens are removed, which ran for three of five regions and excluded 49 cells in Muğla, 16 in North Evia and 61 in Montiferru, with none arising in Manavgat or Bejís, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.



**Figure 2: Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed  $\sim 510$  m analysis grid ( $17 \times 17$  pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

### 3.3. Burned-landcover admissibility gate

215 Before any modelling each region passes a gate asking what fraction of  
its burned cells is dominated by natural vegetation, using ESA WorldCover  
classes (Zanaga et al., 2022) on the same cells. A region is admitted at 0.50  
with at least 30 burned cells, and rejected as a cropland-dominated control  
at 0.50 cropland. Verdicts and the purpose of the gate are in Section 4.1, the  
220 full rule in Appendix C.1.

### 3.4. Predictor variables

Two feature sets are used. The **baseline** is terrain, fuel and green-  
ness: elevation, slope, dominant land cover and a predictor-window median  
vegetation-index composite, all static or near-static over the timescale at  
225 which fire danger varies. The **thermal** set adds six pre-fire channels: current  
land surface temperature, its anomaly against a four-year window-symmetric  
baseline at the same cell, the Temperature-Vegetation Dryness Index and its  
difference against that baseline, and two coordinate-informed products, a  
downscaled and a fused surface temperature. The two differenced channels  
230 are the ones constructed to isolate the dynamic anomaly. TVDI’s wet and  
dry edges are percentiles of the values a given area and window happen to  
contain, so **it is not portable as a physical quantity independently of  
any concept shift** (Appendices C.4, C.5).

### 3.5. Cell aggregation, validity and analysis populations

235 Cell-level values are means over the  $\sim 510$  m cell from valid 30 m pixels  
only, with the valid fraction recorded. A cell is **valid\_for\_modeling** when  
it is analysis-eligible, meaning not excluded for pre-label burning, and its

predictors are valid: joint finite NDVI, elevation and slope support over at least 30 % of the cell, finite means for those three channels, and at least  
 240 one valid land-cover pixel. **Thermal completeness is not part of the definition:** depending on the region, 6 % to 58 % of valid cells carry at least one missing thermal channel, and missing values are imputed inside the fitting pipeline (Section 3.6) rather than by excluding the cell. The **primary** population is natural vegetation, cells whose combined tree, shrub and grass  
 245 fraction reaches 0.50, which excludes cropland from every burnable mask. The **secondary** population is all valid cells; the frozen export carries it for the within-region arm in two regions, reported as a sensitivity in Appendix A(v), and the transfer matrix is defined on the primary population only.

### 3.6. Classifier

250 The two feature sets of Section 3.4 are nested, the thermal set being the baseline plus the six thermal channels. Land cover is one-hot encoded. Missing numeric values are median-imputed and the categorical channel most-frequent-imputed, with the imputers fitted inside each training fold, and on the source alone in transfer, so no held-out statistic enters a fit in the raw  
 255 arms. Features are not otherwise standardised. Standardisation appears only in the adapted arms of Section 3.9, where target feature statistics, never target labels, enter the transform by design, and missing values there are filled with the region’s own mean. The classifier is a random forest (Breiman, 2001) with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced  
 260 class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison here is confounded by a model choice.

### 3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Cell  $i$  at grid position  $(r_i, c_i)$ , from  
 265 `row_500m` and `col_500m`, is assigned to the block

$$b_k(i) = (\lfloor r_i/k \rfloor, \lfloor c_i/k \rfloor). \quad (1)$$

Five folds are drawn over whole blocks, stratified by label, so no block is split between training and test. The block size  $k$  is reported at 2, 10 and 20 cells, about 1, 5 and 10 km.

Every score is a ROC-AUC on a named set of cells  $F$ , its **evaluation**  
 270 **frame**. With  $F^+$  the burned and  $F^-$  the unburned cells of  $F$ , and  $s_i$  the predicted score,

$$\text{AUC}(s; F) = \frac{1}{|F^+| |F^-|} \sum_{i \in F^+} \sum_{j \in F^-} \left[ \mathbf{1}(s_i > s_j) + \frac{1}{2} \mathbf{1}(s_i = s_j) \right]. \quad (2)$$

This is the probability that a burned cell in  $F$  outranks an unburned cell in  $F$ . Changing the frame therefore changes the metric even when no score changes (Section 3.12).

275 Uncertainty is a spatial-block bootstrap. Let  $\beta_1, \dots, \beta_M$  be the blocks of (1) holding cells of  $F$ . Replicate  $b$  draws  $M$  indices  $u_{bm}$  uniformly with replacement:

$$F^{*b} = \biguplus_{m=1}^M \beta_{u_{bm}}, \quad \text{CI}_{95} = \left[ Q_{0.025} \{ \theta(F^{*b}) \}_b, Q_{0.975} \{ \theta(F^{*b}) \}_b \right]. \quad (3)$$

Here  $\theta$  is the statistic and  $Q$  the 2.5 and 97.5 percentiles over 1000 replicates, seed 42. A single-class replicate is discarded. Differences are formed  
 280 within each replicate, so they are paired. The block size of each interval is stated with the result. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an  
 285 interval.

The twenty transfer directions are not independent, since each region appears in eight. The **pair-cluster bootstrap** therefore resamples the ten unordered region pairs, each carrying its set  $\pi_p$  of ordered directions, and averages the carried values  $\delta_{st}$ :

$$\bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t) \in \pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\pi_{u_{bm}}|}. \quad (4)$$

290 The interval is the same percentile form, over 20,000 replicates for Section 4.4. Clustering by target region replaces  $\pi_p$  by the four directions sharing a target. A quantity with one value per held-out scar or target region gets a Student t interval over those  $n$  units,

$$\bar{x} \pm t_{0.975, n-1} s_x / \sqrt{n}. \quad (5)$$

The effective sample is thus ten pairs or five regions for  
 295 direction-level intervals, and at most eight scars from four regions for scar-level ones.

### 3.8. Cross-region transfer protocol

For each ordered pair the model is fitted on the source population and applied to the target with **no target label used at any point**: no refitting,  
 300 no threshold selection, no calibration. Both feature sets are transferred, so the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions are computed.

### 3.9. Label-blind domain adaptation

305 Two label-free remedies are tested on every direction, in both feature sets.

**Region-wise z-score.** Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. For feature  $j$  in region  $R$ ,

$$z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \quad \mu_j^R = \frac{1}{n_j^R} \sum_{i \in O_j^R} x_{ij}, \quad \sigma_j^R = \left( \frac{1}{n_j^R} \sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2 \right)^{1/2}, \quad (6)$$

where  $O_j^R$  holds the  $n_j^R$  cells with an observed value (ddof 0). A missing  
 310 value is first set to  $\mu_j^R$ , so it becomes zero, and  $\sigma_j^R < 10^{-12}$  is replaced by 1. Land cover is not transformed. Target feature statistics, never target labels, thus enter the adapted arms by design. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

315 **CORAL after region-wise z-score.** The source covariance is aligned to the target’s by the standard whitening-recolouring map (Sun et al., 2016). With  $Z_R$  the  $n_R \times d$  matrix of z-scored numeric features, one row per cell,

$$Z_s^{\text{al}} = Z_s (C_s + \lambda I)^{-1/2} (C_t + \lambda I)^{1/2}, \quad C_R = \frac{1}{n_R} \sum_{i=1}^{n_R} (z_i - \bar{z}_R)(z_i - \bar{z}_R)^\top, \quad (7)$$

with  $\lambda = 10^{-5}$  (ddof 0). Both means are zero after (6), so the general map’s mean terms vanish. Matrix powers use a symmetric eigendecomposition, eigenvalues floored at  $10^{-12}$ . Critically **the transform is applied to the source only**; the target stays at  $Z_t$ , and the classifier is refitted on  $Z_s^{\text{al}}$ . Neither variant sees a target label, and both are verified label-blind at run time.  $\lambda$  sensitivity was assessed over nine values on four of the twenty directions, moving transfer AUC by at most 0.014, and no value of  $\lambda$  was selected on performance; the  $\lambda = 1$  of the original CORAL formulation lies outside that sweep, while the value used throughout remains  $\lambda = 10^{-5}$  (Appendix A(b)).

### 3.10. Transfer-gap decomposition and the concept-shift criterion

For a direction into a target, let  $A_w$  be the target’s within-region thermal AUC under blocked cross-validation,  $A_{\text{raw}}$  the raw transfer AUC and  $A_{\text{ad}}$  the transfer AUC after adaptation method  $m$ , all on the same target cells. The gap is decomposed as

$$G = A_w - A_{\text{raw}}, \quad R_m = A_{\text{ad}} - A_{\text{raw}}, \quad U_m = A_w - A_{\text{ad}}, \quad \rho_m = R_m/G, \quad (8)$$

so  $R_m + U_m = G$ . The recovered fraction  $\rho_m$  is signed and unclipped. **When adaptation lowers AUC,  $R_m$  and  $\rho_m$  are negative and re-**

335 **ported as negative recovery, never set to zero.** No fraction is reported when  $G \leq 0$ . Intervals come from the paired bootstrap of (3) on 2-cell target blocks, resampling within-region out-of-fold and transfer scores together and evaluating (8) per replicate; replicates with  $|G| < 10^{-6}$  are dropped. Where one method is shown per direction it is the one with the higher  $A_{\text{ad}}$ , a choice  
 340 that uses target labels. Section 4.3 shows the remainder should not be read as a conditional residual, because much of it is incurred inside a single region (Appendices A(j), C.7).

The mechanism is diagnosed by **signed univariate association**: the raw ROC-AUC of each numeric predictor against **burned** is computed per  
 345 region and never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 is a direction rather than weakness. A reversal is called bootstrap-supported only when the two regions' point estimates fall on opposite sides of 0.5 **and each region's own interval excludes 0.5**. That is stricter than requiring the intervals to be disjoint, and a feature failing the second condition is recorded  
 350 as a point reversal only.

### 3.11. Interventions

**Pooled multi-region training.** Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks  
 355 whether pooling recovers what single-source transfer loses.

**Removal of direction-reversing features.** The two predictors reversing with bootstrap support **on the frames as drawn** are **elevation\_mean** and **lst\_anomaly\_mean**. Section 4.4 withdraws that support under an equalised frame, so the selection rule is frame-dependent and this arm is

360 reported as a measurement under the original protocol, not as a consequence  
of an established reversal. Both are dropped and everything refitted, within-  
region and across every direction, and each is dropped singly so the cost can  
be attributed. The two features are selected by the same reversal analysis  
the result is then read against, so **both quantities are post-selection**  
365 **estimates** with no correction applied.

### 3.12. Evaluation frames and controls on the transfer path

Let  $V_R$  be the primary population of region  $R$  and  $P_R \subset V_R$  its burned  
cells. By (2), a frame fixes which burned and unburned cells are compared.  
The **region-wide frame** is  $V_R$ , the rectangle as drawn, used in Sections 4.2  
370 and 4.5.

**Scar frames.** A scar is an 8-connected component  $K$  of  $P_R$  with at least  
50 cells. Its **scar + 2 km collar** is

$$H_K = \left\{ i \in V_R : \min_{j \in K} (|r_i - r_j| + |c_i - c_j|) \leq m \right\}, \quad (9)$$

with  $m = \text{round}(2/0.45) = 4$  grid steps, computed as  $m$  steps of 4-  
connected dilation. Its negatives are all fire-adjacent. Four evaluations  
375 use it (Table 2). A scores out-of-fold predictions from 5-fold blocked cross-  
validation ( $k = 10$ ) on  $V_R$ , and B scores the same predictions on  $H_K$ . B is  
thus the **same blocked model restricted** to the scar area, which isolates  
the evaluation region from the training regime. C is **leave-one-scar-out**: a  
model fitted on  $V_R \setminus H_K$  is scored on  $H_K$ , skipped if either set has one class. D,  
380 the **foreign-region evaluation**, averages over the four other regions a model  
fitted on that region’s population and scored on  $H_K$ . Buffers of 2, 5 and 10

km were run, 2 km primary. Three controls reuse A’s predictions, each averaged over 20 draws without replacement. The **prevalence-matched** control draws  $|H_K^+|$  cells from  $P_R$  and  $|H_K^-|$  from  $V_R \setminus P_R$ . The **negative-pool** control keeps the drawn burned cells but uses the scar’s own negatives  $H_K^-$ , and the positive-pool control does the converse. A **within-region half-split** (modelled cells cut at the median of a grid axis, both axes and directions, a split discarded when either half is single-class) completes the set, under the transfer protocol of Section 3.8 unchanged. The per-split and per-scar positive counts are unequal and bear on the interpretation (Appendix A(i)).

**Distance collars.** With 0.45 km per grid step on both axes, the distance to the nearest burned cell and the collar of radius  $r$ , for  $r$  of 5 and 10 km, are

$$d_i = 0.45 \min_{j \in P_R} \|(r_i, c_i) - (r_j, c_j)\|_2, \quad F_R(r) = \{i \in V_R : d_i \leq r\}. \quad (10)$$

Every burned cell has  $d_i = 0$  and is kept, so only far-field negatives leave. In transfer the model is fitted on  $F_s(r_s)$  and scored on  $F_t(r_t)$ , with the pairs of Table 3 and  $r = \infty$  for the region-wide frame.

**Every collar frame,  $H_K$  and  $F_R(r)$  alike, is defined from burned cells, so it is label-conditioned.** It is a diagnostic of how the scoring extent shapes the metric. It cannot be drawn before a fire.

### 3.13. Leakage control and reproducibility

An explicit forbidden-column set is enforced at every model fit as an assertion rather than a convention: coordinates and their normalised forms,

every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets. The natural-vegetation mask defines the population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates, with one qualification: the diagnostic bootstraps of the released appendices use per-measure offsets from that seed rather than the seed itself, so that independent measures do not share a resampling draw. Two of the twenty transfer verdicts are not stable across seeds and are identified in Section 4.5. Because the transfer analysis runs in an environment separate from the upstream pipeline’s, every within-region model was refitted against the frozen upstream output: the within-region comparisons agree exactly and the twenty transfer directions to within  $1.6 \times 10^{-7}$ , with the tolerance that applies if the library version is not pinned in Appendix C.5(vi). Headline results are repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four classifier capacities, and where a conclusion depends on one of those choices **the dependence is reported rather than resolved by choosing the favourable setting** (Appendices A, C.6).

### 3.14. Same-geography event-to-event comparison (*Muğla 2021 versus 2022*)

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell bootstrap used elsewhere. **Season, year and population all differ**, since the 2022 arm is defined by removing the 2021 scar (Appendices A(m), C.3).

## 4. Results

### 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate and the negative control fails it as intended: Kozan 2023 returns a natural-vegetation  
430 fraction of 0.017 against the 0.50 threshold and is excluded. The separation is not marginal, the five admitted regions carrying 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating natural-fuel combustion  
435 from post-harvest stubble burning, which MCD64A1 does not distinguish, **though one negative control cannot establish it**. North Evia is analysed on an extended AOI, which leaves the burned scar essentially unchanged while cutting TSG prevalence from 0.676 to 0.287; the effect on transfer is in Appendix A(a).

### 4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region, the increment’s bootstrap interval  
excluding zero in all five at 1 km and at 5 km blocking (Fig. 3) — the two scales this design supports as intervals. At 10 km the point estimates hold,  
445 from +0.048 to +0.154, but rest on 6 to 33 positive-carrying blocks and are indicative.

*Table note (resampling units).* The bootstrap resamples spatial blocks, so an interval’s reliability is bounded by the number of blocks carrying at least one burned cell, and those counts fall from 192 to 843 at 2 cells to **6**

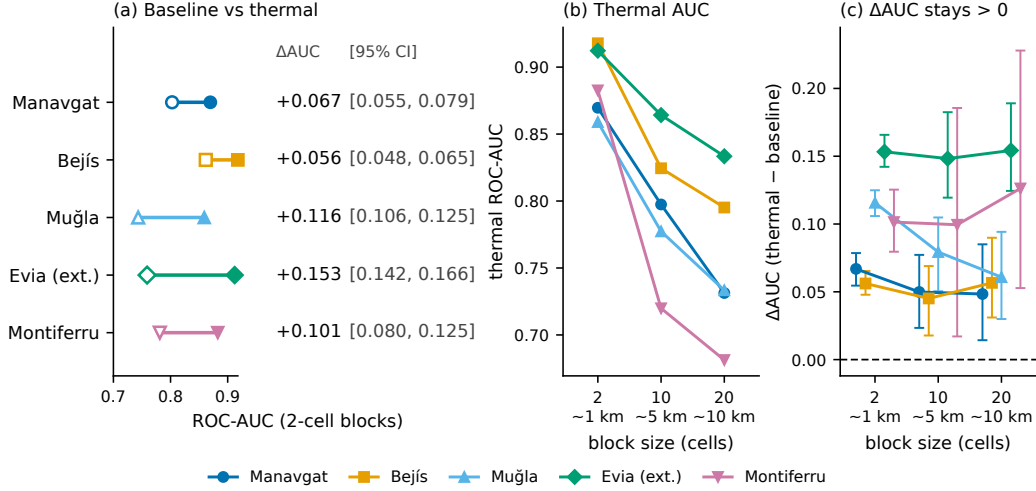


Figure 3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell ( $\sim 1$  km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired  $\Delta AUC$  and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells,  $\approx 1$ , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment  $\Delta AUC$  against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 1, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes (2, 10, 20 cells)**, and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells  $\approx$  1/5/10 km. Baseline and Thermal columns are ROC-AUC; the 95 % CI belongs to  $\Delta$ AUC.

| Region                    | Block | Baseline | Thermal | $\Delta$ AUC | 95% CI           |
|---------------------------|-------|----------|---------|--------------|------------------|
| Manavgat 2021             | 2     | 0.803    | 0.870   | +0.067       | [+0.055, +0.079] |
|                           | 10    | 0.748    | 0.797   | +0.050       | [+0.023, +0.077] |
|                           | 20    | 0.683    | 0.731   | +0.048       | [+0.014, +0.085] |
| Bejís 2022                | 2     | 0.862    | 0.918   | +0.056       | [+0.048, +0.065] |
|                           | 10    | 0.779    | 0.825   | +0.045       | [+0.018, +0.069] |
|                           | 20    | 0.739    | 0.795   | +0.057       | [+0.031, +0.090] |
| Muğla 2021                | 2     | 0.743    | 0.859   | +0.116       | [+0.106, +0.125] |
|                           | 10    | 0.698    | 0.777   | +0.079       | [+0.050, +0.105] |
|                           | 20    | 0.673    | 0.733   | +0.061       | [+0.030, +0.094] |
| North Evia 2021<br>(ext.) | 2     | 0.759    | 0.912   | +0.153       | [+0.142, +0.166] |
|                           | 10    | 0.716    | 0.864   | +0.148       | [+0.119, +0.183] |
|                           | 20    | 0.679    | 0.833   | +0.154       | [+0.124, +0.189] |
| Montiferru 2021           | 2     | 0.781    | 0.883   | +0.101       | [+0.080, +0.125] |
|                           | 10    | 0.620    | 0.720   | +0.100       | [+0.017, +0.186] |
|                           | 20    | 0.555    | 0.681   | +0.126       | [+0.053, +0.228] |

450 **to 33** at 20 cells. An interval built on six such blocks has no meaningful coverage, so **the 20-cell row should be read as indicative rather than as an interval**. The 10-cell row is the coarsest blocking this design supports properly, every region there having 16 to 70 positive-carrying blocks, and the increment holds at that scale in all five regions (Appendix A(c)).

455 This within-region result is not itself novel; it is reported because the transfer arms below are measured against it.

#### *4.3. What the evaluation frame is worth, within one region*

This section isolates the effect Section 4.4 applies between regions. It comes first because it is measured on one model in one region with no transfer  
460 involved, so nothing the transfer arms do can explain it (Appendix A(i)).

**Where the skill is lost, on a matched comparison.** Four evaluations are reported; the last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first shows why an unmatched comparison misleads. The held-out unit is a burned connected  
465 component of at least 50 cells with all cells within 2 km of it. What makes that harder than a whole region is the composition of its negatives, not its burned fraction, and that is measured rather than argued. Scoring the same predictions on a random sample of region cells drawn at the scar area's own burned fraction gives **0.782 against the region-wide 0.782**, so matching  
470 prevalence changes nothing, at  $-0.000 [-0.003, +0.002]$ ; scoring them on the scar area gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. That swaps both pools at once, so each was swapped singly: replacing only the negatives costs **0.147 [0.104, 0.191]**, the whole of it, and replacing only the positives costs **0.002 [-0.052, +0.048]**, null on average though the per-scar effect runs

475 from  $-0.12$  to  $+0.12$ . **The effect is the negative pool** — every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far field — and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions (`pool_decomposition.json`).

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student  $t$  intervals are over the eight held-out scars.

| Evaluation                        | Model trained on                 | Scored on        | Mean AUC | 95 % CI        |
|-----------------------------------|----------------------------------|------------------|----------|----------------|
| A. Blocked cross-validation, 5 km | the region, scar included        | the whole region | 0.776    | [0.738, 0.814] |
| B. Same blocked model, restricted | the region, <b>scar included</b> | the scar area    | 0.634    | [0.552, 0.716] |
| C. Leave-one-scar-out             | the region, <b>scar withheld</b> | the scar area    | 0.552    | [0.501, 0.602] |
| D. Foreign region                 | another region, 306 to 2,802 km  | the scar area    | 0.555    | [0.495, 0.616] |

480 All four rows are means over the **same eight scars**, and Bejís has no row C because it is its region’s only component of any size, which is why the controls above are computed on nine and report 0.782 and 0.627 against this table’s 0.776 and 0.634. The intervals are Student  $t$  over the scars, this arm’s resampling unit rather than the spatial-block bootstrap used elsewhere.

485 Four scars are in Muğla and two in Montiferru, and **row A is a region-level quantity repeated identically across a region’s scars, so its interval is pseudo-replicated and should not be read as coverage**. That repetition propagates into  $A - B$  and  $A - C$ ; clustering by region gives  **$+0.155$  [ $+0.082$ ,  $+0.228$ ]** and  **$+0.251$  [ $+0.093$ ,  $+0.409$ ]**, both still

490 excluding zero but the second 2.3 times wider, so **the scar-level intervals**  
**on these two differences overstate precision** (Appendix A(i)).

The same model, scored two ways on the same region, differs by  
**0.143 AUC**. Rows A and B are one model and one set of out-of-fold predic-  
tions; the only change is which cells they are scored on. That change alone  
495 costs **0.143** [**+0.077, +0.208**] of the 0.225 [+0.157, +0.293] fall from A to  
C, about two thirds — a ratio of two estimates, compatible with anything  
from roughly a third to nine tenths, so the point is the size of the numerator  
rather than the precision of the fraction. It is the size of the predictor-block  
increments this design itself measures, so a region-wide blocked figure is an  
500 upper bound on what the same model achieves where the fire actually is.

**Neither withholding the fire nor moving 2,800 km has a measur-**  
**able cost**. B minus C, the fire-specific residual on identical cells, is **+0.082**  
[**-0.011, +0.175**]; C minus D, the effect of replacing a same-region model  
with one fitted 306 to 2,802 km away, is **-0.003** [**-0.075, +0.069**]. Both  
505 span zero and both arms sit close to chance, so **with eight scars, three of**  
**them starved of positives, this design cannot establish a fire-specific**  
**residual, only bound it at about 0.18**. It is not shown to be zero on an  
unseen fire; what this design establishes is that it is not established there.  
Nor does it establish the fire *event* as the unit, since the held-out patch is  
510 defined by the labels and its identity cannot be separated from its location.  
Appendix A(z) reports the spread behind row D and the sweeps showing the  
patch definition does not drive the result.

**The increment declines with the holdout, and is not established**  
**once the fire is withheld**. Contribution 2 concerns the paired thermal-

515 minus-baseline difference, so the same evaluations were run on it: +0.056 to  
+0.153 under blocked cross-validation with every interval above zero, +0.027  
on a within-region half-split, **+0.022** [**-0.032**, **+0.077**] under leave-one-  
scar-out, and +0.004 [-0.028, +0.036] across regions. The point estimate  
falls monotonically as the holdout hardens and the last two intervals span  
520 zero, so **the within-region increment is substantially a property of  
interleaved holdout.**

#### *4.4. The same effect between regions, applied to our own matrix*

Section 4.3's effect applies with equal force between regions. This test is  
reported before the transfer matrix because it changes what Section 4.5 and  
525 Appendix A(s) can claim; it withdraws nothing in Section 4.3, which is what  
it is built from. Sources are `aoi_frame_auc_frozen_mugla.csv` and `aoi_  
frame_transfer_frozen_mugla.csv`, which supersede the pre-correction  
files for the reason given below, with `collar_frame_bootstrap.csv` and  
`diagnostics_collar_frame.csv`; code is under `paper/code/` and the  
530 elaboration in Appendix A(w).

**The five areas of interest are not comparable frames.** Each is a  
rectangle drawn around a fire, and they differ by an order of magnitude in  
how much unburnt far field they enclose: the share of modelled cells beyond  
10 km of any burned cell runs from **2.1** % in Montiferru to **63.1** % in Bejís  
535 (Appendix B, Table B6). That far field is not neutral — in Manavgat the  
median elevation of modelled cells rises from 472 m within 5 km of the fire  
to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

**Under an equalised frame the sign reversals do not survive.** Re-  
stricting every region to cells within 10 km of any burned cell removes only

540 far-field negatives — every burned cell is at distance zero and is retained  
at any radius — so the protection against a flattering radius is the sweep,  
not the retention of positives. On that frame all five regions agree in sign  
on elevation, LST and TVDI, and both bootstrap-supported elevation re-  
versals of Table B3 disappear. Under this paper’s own criterion from Sec-  
545 tion 3.10, **no between-region reversal remains bootstrap-supported:**  
the two features straddling 0.5 at the point estimate are the internally differ-  
enced channels, and in each at least one region’s interval covers 0.5 — Evia’s  
by 0.003. The weaker difference instrument that Table B3’s note commits  
this paper to does support four opposite-sided pairs on `lst_anomaly_mean`,  
550 with three caveats that keep it weak. **The honest statement is that no  
between-region reversal meets the strict criterion under the collar,  
and that the LST anomaly differs between regions on the weaker  
one.** Two scope limits belong with it, both in Appendix A(l) and A(w): the  
interval criterion was evaluated at the 10 km collar only, though at the point  
555 estimate the two radii agree exactly; and the collar reduces two regions’ cell  
counts substantially, so part of the loss of support is a loss of power, against  
which features supported in both regions *rise* from 1.20 to 3.40 per direction.

**The sign they now agree on is not the one the dryness framing  
predicts:** at the point estimates a hotter pre-fire surface burned *less* in every  
560 region, and on mutual adjustment temperature survives where greenness does  
not, so the absolute channels behave here as static land-surface descriptors  
rather than as dryness. Within distance to the nearest burned cell the sign  
crosses 0.5 in Manavgat and is attenuated to near-null in Montiferru, so in  
those two it is largely the residual spatial gradient the collar was introduced

565 to remove. The stratifications, the correlations and the reciprocal adjustment are in Appendix A(k).

**The same test removes a result of our own.** The diagnostic that best ordered transfer in our matrix is built from exactly the signed AUCs shown above to be frame artefacts, and was correlated against transfer on 570 the same unequal frames. Recomputed under the collar it is unanimous — 1.0 in all eighteen directions, variance exactly zero — so it has no variance left to correlate, and its cosine variant falls from  $\rho = +0.81$  to  $-0.06$ . The correction does not weaken that result; it dissolves it (`diagnostics_collar_frame.csv`, `collar_increment_and_cosine.csv`).

575 **Two further arms move with the frame.** The same-geography arm is the most extreme case in the cohort — a fixed study area is not a fixed evaluation frame, and its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021 — so under the collar its two arms fall on the same side of 0.5 on elevation, the feature whose reversal we had 580 reported. The claim is scoped: that arm is reconstructed from the 2021 predictors, valid for the year-invariant channels and **not** the seasonal ones, so **only elevation and slope carry a verdict, on both the reversal is removed, and on the thermal channels the arm is silent**; it rests on eleven positive-carrying blocks against a floor of sixteen (Appendices A(m), 585 A(o)). The within-region increment, by contrast, **survives**, positive in all five regions at a mean of  $+0.077$  against  $+0.086$  as drawn.

**The transfer matrix moves as well.** Restricting source and target to the same collar:

**A data-provenance defect in this arm was found and corrected:**

Table 3: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use a 10-cell ( $\approx 5$  km) spatial-block bootstrap on the target, 1000 replicates, seed 42; **Table B9 reports the same matrix under 2-cell ( $\approx 1$  km) blocking**, which is why its supported counts are the larger 12 and 6 (Section 4.5). Per-direction bounds are in `aoi_frame_transfer_frozen_mugla`.

**CSV.**

| Source frame | Target frame             | Mean target AUC | Above chance    | Below chance | Supported above / below | Paired thermal delta |
|--------------|--------------------------|-----------------|-----------------|--------------|-------------------------|----------------------|
| full         | full ( <b>Table B9</b> ) | 0.541           | 14 of 20        | <b>6</b>     | 9 / <b>4</b>            | +0.004               |
| full         | 10 km                    | 0.576           | 17 of 20        | 3            | 11 / 1                  | +0.003               |
| 10 km        | full                     | 0.570           | 17 of 20        | 3            | 11 / 3                  | +0.014               |
| <b>10 km</b> | <b>10 km</b>             | <b>0.616</b>    | <b>19 of 20</b> | <b>1</b>     | <b>15 / 1</b>           | <b>+0.023</b>        |
| 5 km         | 5 km                     | 0.608           | 19 of 20        | 1            | 12 / 0                  | +0.014               |

590 one region’s predictor file at the canonical path had come to differ from the  
one the frozen tables were computed on, so every arm here was re-run against  
the frozen export. It moves forty of a hundred per-direction values by up to  
0.022 and leaves **every headline quantity above unchanged to within**  
**0.0012**, with the signed AUCs of that region’s two channels moving by at  
595 most 0.008 and no verdict changing (Appendix A(w)). The table reports the  
corrected values, and reproduces the frozen matrix at 0.541 against Table  
B9’s 0.541 and 14 of 20 exactly. **The baseline control must be restated**  
**on this frame**: the static baseline transfers at 0.593 against 0.616, a paired  
difference of +0.023 rather than +0.004, so the control holds in kind but the  
600 gap is about six times larger once frames are comparable, and about four  
times at the 5 km collar.

**What this settles, and what it leaves standing.** Five quantities  
reported below are properties of the frames rather than of the predictor-  
burning relationship, and are identified as such where they appear: the count  
605 of six anti-predictive directions, which becomes one; the sign reversal of ele-  
vation, LST and TVDI as a mechanism; the sign-agreement diagnostic; the  
same-geography arm; and the paired thermal contribution, +0.004 as drawn

against +0.023 equalised. That last quantity carries the portability null, so it is given an interval on the frame this section argues for: **+0.023** [−**0.004**,  
610 **+0.048**] under the pair-cluster resampling of Appendix A(o). It still spans zero, so the null survives the correction — but only just, and under the alternative admissible unit, clustering by target region, it does not (Appendix A(w)). What survives is the central negative result, and its size must be stated on a matched comparison: setting 0.616 against a within-region refer-  
615 ence of about 0.87 would compare a collar number with a full-rectangle one at 1 km blocking. Recomputed on the same frame at 5 km blocking that reference is 0.772, so the shortfall is **+0.155** [**+0.094**, **+0.217**], a Student  $t$  interval over the five target regions rather than the spatial-block bootstrap used elsewhere, and numerically almost identical to Section 4.3’s negative-  
620 pool effect over nine scars, which is a different quantity on a different unit. It is real, and 0.155 rather than the 0.25 an unmatched comparison implies. Appendix C.5(ix) records the frame as a limitation of this cohort rather than of the method.

#### *4.5. Cross-region transfer, and what label-free adaptation does to it*

**Everything in this section is computed on the frames as drawn**  
625 **and should be read against Section 4.4**, which has shown the frames are not comparable and that the within-region reference used here is itself frame-dependent. The as-drawn matrix is reported because it is what the original analysis protocol yields; per-direction values are in Appendix B, Table B9,  
630 and the supporting arms in Appendix A(ix).

**Raw transfer is heterogeneous and includes anti-predictive directions.** Target AUC spans 0.326 to 0.686, and Fig. 4 gives the matrix

direction by direction. The support counts depend on the blocking scale, so both are reported: at 2-cell ( $\approx 1$  km) blocking twelve of twenty directions  
635 are above chance with interval support and six below; at the 10-cell ( $\approx 5$  km) blocking this design defends, nine are above and four below with seven uncertain, the counts Table 3 uses. Of the below-chance directions only Manavgat to Bejís survives frame equalisation, and two verdicts are not stable across bootstrap seeds (`transfer_ci_blocksize.csv`). Even the best raw  
640 transfer sits far below that target’s own within-region 0.870, the raw deficit running 0.184 to 0.592 against a reference Section 4.3 shows is not matched to a transfer evaluation.

**In precision terms it is worse than the ROC figures suggest.** A susceptibility surface is used as a ranked area budget, so precision-recall is the  
645 operational quantity. PR-AUC averages **0.156 against a no-skill baseline of 0.136**, and **six of twenty directions fall below their own baseline at the point estimate**, five of them with intervals entirely below it; the exception is Bejís to Manavgat, whose interval covers its baseline. Only one direction exceeds twice its baseline. These are frame-as-drawn quantities  
650 and the PR arm was not recomputed on the collar (Appendix B, Table B4).

**The static baseline does not transfer either**, at a mean of **0.537** against **0.541** for the thermal model, so the failure is not specific to the dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel. Section 4.4 restates this control on the  
655 equalised frame, where it is 0.593 against 0.616.

**The thermal block’s paired contribution is +0.004, and the mean is not the informative statistic.** The individual contributions

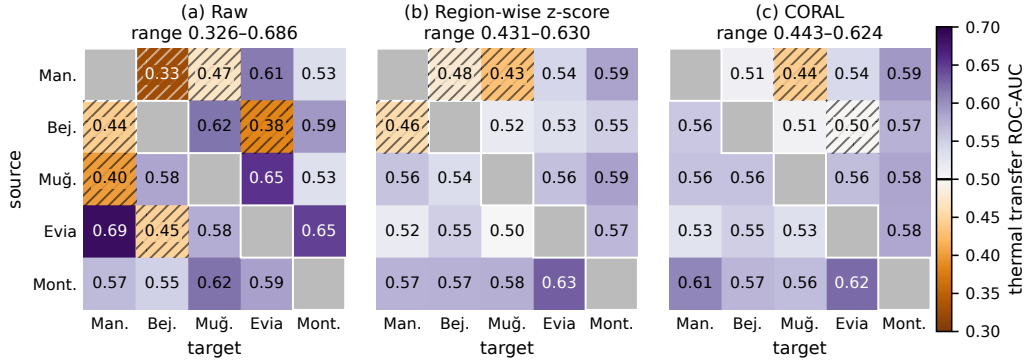


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source-target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table B9.

span **-0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from  $[-0.027, +0.034]$  to  $[-0.037, +0.046]$ , and none propagates within-direction sampling variability. The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

**Label-blind adaptation compresses the matrix toward chance rather than repairing it** (Fig. 5). Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, and fourteen of twenty move closer to chance — which helps the directions that failed and harms those that worked. The committed-in-advance CORAL arm averages **0.552**; taking whichever method scores better per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is **an oracle upper bound rather than an achievable result**. Even the oracle only reaches the reference a model can reach on an unseen scar (Section 4.3), so alignment is regressing the matrix onto that reference rather than failing beneath it. What it cannot do is exceed the reference, and a sign reversal is not a distribution mismatch that realigning inputs would repair. Appendix A(ix) reports the recovery fractions, including the seven directions with *negative* recovery.

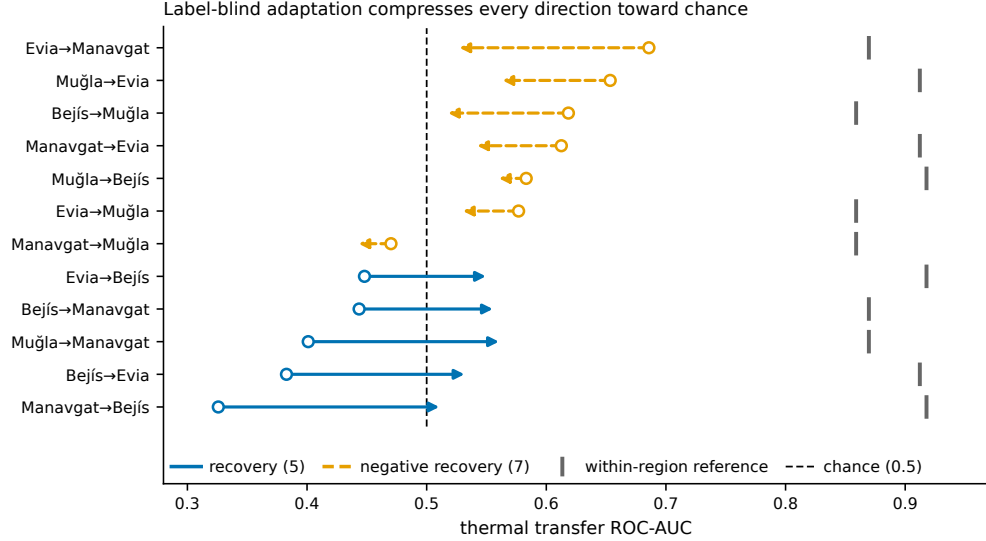


Figure 5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves away from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of  $-0.862$ . No arrow reaches its within-region tick.

#### 4.6. *Further arms*

Six further arms bear on the findings above without changing them: the contrast pair (A(s)), the two interventions (A(n)), the sensitivity summary (A(v)), the same-geography two-event arm (A(m)), the distance curve (A(t)) and the label-budget curve (A(u)). Each is stated there with its own limits. Two are plotted here because the shape of the result is the argument: pooling every other region never beats the best single source for any target (Fig. 7); removing the direction-reversing features costs within-region skill and returns nothing measurable on transfer (Fig. 6). The two that bear directly on Sections 5.3 and 5.6 are the interventions, where a local cost of  $-0.081$  is measured against a transfer return of  $+0.014$   $[-0.017, +0.045]$  whose interval spans zero, and the label budget, where thirty-two labelled 5 km blocks recover 85 to 89 % of the target's matched ceiling in three of six directions and 30 to 57 % in the rest — 7 to 20 % of the target's population, drawn from the event being predicted, which is not a resource available before that event burns.

## 5. Discussion

### 5.1. *Reading the two findings together*

Section 1.3 states the two findings and Section 4 establishes them; this section argues from them. The relation between them is what makes the paper cohere: the first is not a caveat attached to the second, it is the instrument that sets its size. Applied to our own matrix it withdrew five

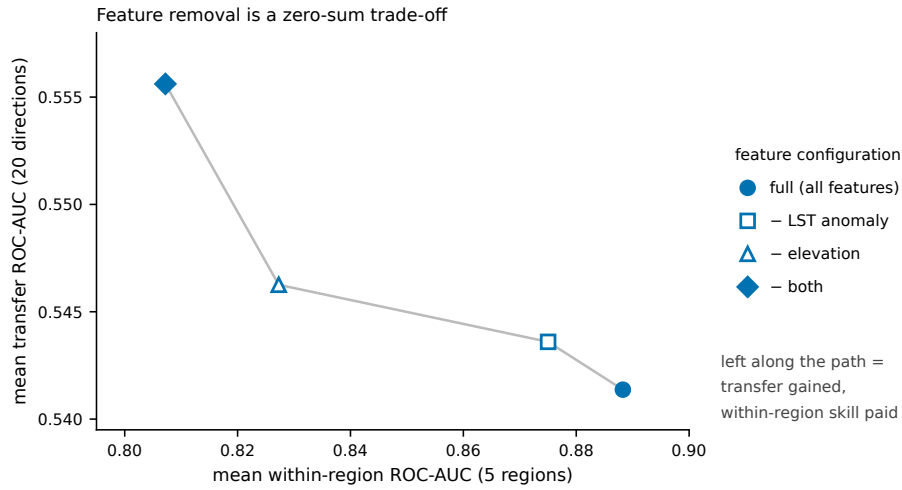


Figure 6: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ( $0.541 \rightarrow 0.556$ ) while within-region skill falls monotonically ( $0.888 \rightarrow 0.807$ ). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

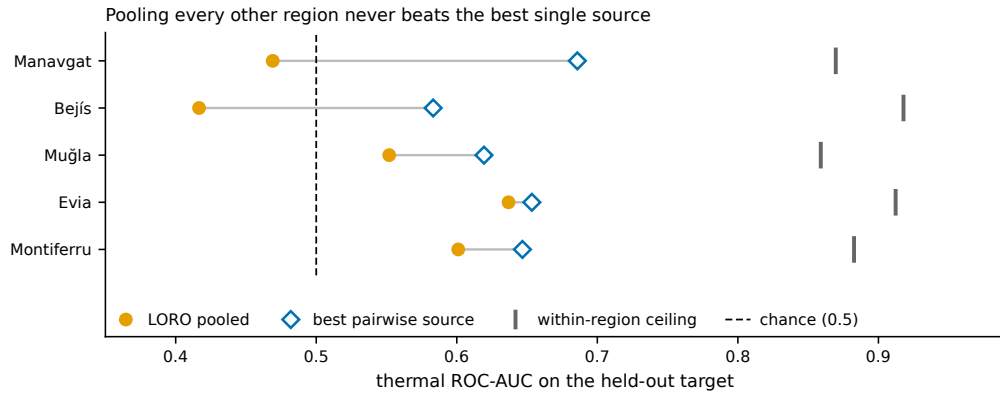


Figure 7: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

claims, and what it left standing is a shortfall in transferred skill, not a reversed relationship.

705 *5.2. Why the thermal increment is real but local*

The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference is already present inside a single region. The increment is local in a specific sense: it holds where held-out cells are interleaved with training cells and most of it  
710 is gone once they are not, before the fire or the region changes. It is not an artefact to be explained away, surviving every sensitivity arm of Appendix A(a)–(h), but it is established under interleaved validation and not beyond it — and even there the window-closure arm **weakens monotonically in Evia**, so what holds everywhere is survival, not improvement.

715 The natural objection is that Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not answer it:** Section 4.4 shows it is the most extreme frame artefact in the  
720 cohort. Three further confounds were never resolved in any case: season and year, which **cannot be resolved in this study area**, an unfixed population, and a positive-block count below this design’s own floor (Appendix C.5(ii)). With one fire per region everywhere else and that arm withdrawn, **this cohort provides no evidence that the transfer shortfall is re-**  
725 **gional rather than event-specific.** Appendix A(t) adds a length scale for the within-region decay but cannot turn it into an attribution either.

One conclusion survives from the other direction: the static baseline

transfers no better than the dynamic block, so whatever the shortfall is, it is not the thermal block's peculiarity.

730 *5.3. What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays well below the within-region ceiling (Appendix A(n)), so it does not manufacture the missing conditional information. Removing the reversing predictors costs  $-0.081$  of within-  
735 region skill with interval support in every region, and returns  $+0.014$   $[-0.017, +0.045]$  in mean transfer.

That pair of numbers is easy to read as an exchange, and it is not one. Both arms are null on the portability axis — the thermal block's own contribution is  $+0.004$ , removal returns  $+0.014$ , both intervals spanning zero —  
740 so they measure a local cost and no compensating gain. That is not a conservation law and not a rate at which local skill can be sold for portability; no such rate is estimated here.

*5.4. The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data con-  
745 firmed the null: the regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: subsampling Muğla, much the largest population, to Manavgat's cell count and separately to its positive count leaves its transfer  
750 behaviour inside the subsampling range in both roles. The error was in the

hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology (Archibald et al., 2013) in general.

### 5.5. *The empirical contrast with Dimarco et al.*

Dimarco et al. (2026) transfer successfully across a comparable Mediter-  
755 ranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours the state of a surface in one season, which suggests the relation between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here:  
760 their response variable is human-driven **ignition**, dominated by access and activity, while ours is burned **area**, dominated by spread, so a predictor-class explanation and a response-variable explanation are confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this  
765 cohort the place-attribute class does not travel either.

### 5.6. *Implications*

In precision-recall terms, which is how a susceptibility surface is used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136 **on the frames as drawn**, six of twenty falling below their own  
770 baseline (Section 4.5). Whatever the ROC figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

The paper supports one concrete change in reporting: alongside a spatially blocked within-region figure, report skill on a held-out burn scar and

its surroundings. On these five regions the two differ by about 0.14 ROC-  
775 AUC on the same model, the size of the effect such papers usually claim, so  
**a blocked figure alone should be read as an upper bound.** Transfer  
skill likewise has to be *measured* on the target region rather than assumed  
from a within-region figure, and where a model must be moved the resource  
that closes the gap is target labels (Appendix A(u)): a real answer, but not  
780 a cheap one.

The results also bound what unsupervised alignment can be asked to do.  
Even the oracle selection, which uses the target labels the protocol forbids,  
only reaches the reference a model can reach on an unseen scar (Section 4.5).  
Alignment is not failing far below an achievable target; it is regressing the  
785 matrix onto it, at the cost of the directions that already worked. A sign  
reversal is not a distribution mismatch that realigning inputs repairs.

### 5.7. Limitations

Ten limitations are stated in full in Appendix C.5; four bind the conclu-  
sions above. **The frames are not comparable and this cohort cannot**  
790 **fully repair it** (Section 4.4): the collar radius is itself a choice — 5 km and  
10 km give 0.608 against 0.616 — and the frames were fixed upstream, so  
we can restrict them but not extend them. Any future cohort should fix the  
frame by an explicit accessible-area rule (Barve et al., 2011) before any pre-  
dictor is computed; that is the main design lesson of this paper. **Each region**  
795 **contributes one fire season**, so regional concept shift is confounded with  
event meteorology and the shortfall cannot be attributed to region rather  
than event (Section 5.2). **The diagnostic correlations rest on an effec-**  
**tive sample of ten region pairs**, so the diagnostic results of Appendix D,

successes and failures alike, read at that power. And **the interval-support**  
800 **counts are less stable than the point estimates behind them:** the  
published split turns on a lower bound of  $-0.00045$ . The remaining six,  
from the absent meteorological covariates to the classifier comparison made  
by point estimate only (Appendix A(h)), bear on scope rather than on the  
conclusions above.

## 805 6. Conclusions

Where a fire model is scored decides what it appears to know. Holding  
model, predictors and fitting fixed and changing only which cells are scored,  
moving from a whole study region to the burn scar and its 2 km collar  
costs **0.143 ROC-AUC**, with a control measuring the cause as the negative  
810 pool's composition rather than class balance. That is the size of the predictor-  
block increments this design itself measures, so a region-wide validation figure  
should be read as an upper bound on what a model achieves where fire  
actually occurs, and reported as one.

That correction is not only other people's problem. Applied to our own  
815 five-region matrix it withdraws five of our claims. The study areas enclose  
very unequal far fields, and equalising them to a 10 km collar lifts mean  
transfer from 0.541 to 0.616, reduces the directions below chance, and leaves  
no sign reversal supported between regions under our own criterion. The  
same correction applies to the one arm that held place fixed, two fires in one  
820 study area eleven months apart, which we had wrongly exempted because  
its geography was constant while its evaluation frame was not: its elevation  
reversal does not survive equalisation either, and that arm can speak only

for the year-invariant channels. What had looked like a reversed predictor-burning relationship was mostly a statement about how five rectangles were  
825 drawn.

What survives is the central result. Pre-fire thermal dryness adds a real and repeatable within-region increment in all five regions, surviving spatial blocking at about 5 km and a predictor window closed up to two weeks before the first labelled burning. It does not travel: on matched frames and blocking,  
830 equalised transfer of 0.616 falls 0.155 short of the within-region reference; the paired cross-region contribution spans zero on both frames under the resampling unit we treat as primary, though not under every admissible one (Section 4.4), with a sign that varies by pair; and the static baseline transfers no better on either frame, so this is not a peculiarity of thermal predictors.  
835 Label-free alignment compresses most directions towards chance rather than repairing them, and supervised recalibration works but is not cheap.

Two limits belong with all of this. The agreed thermal direction is the opposite of the one the dryness framing predicts — hotter pre-fire surfaces burned less in every region, and holding greenness does not remove it while  
840 holding temperature reverses greenness in two regions — so on this cohort the absolute channels behave as static land-surface descriptors rather than as a dryness index. And with one fire per region, and the single arm that held place fixed now withdrawn, this design measures a transfer shortfall but cannot attribute it to region rather than to event.

845 Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions. And fix the

evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both  
850 a transfer failure and a mechanism to explain it. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables.

## **Appendix A. Appendix A. Where the rest of the appendices are**

Two appendices are printed here: **Appendix B**, the diagnostic tables  
855 below, and **Appendix C.5**, the ten limitations. They carry the per-direction numbers and the limitations against which the claims are checked, so the paper can be assessed without leaving it.

**Appendix A and the protocol sections C.1 to C.4, C.6 and C.7 are released with the paper rather than printed in it**, at `paper/supplementary_appendices.md`  
860 in the repository named in the declarations. They keep their section names there, so a pointer such as Appendix A(c) or Appendix C.2 in the text resolves in that document unchanged. Nothing in them is evidence a claim depends on that appears nowhere else; every number in them also sits in a frozen artefact named in the text. The lettering of this  
865 appendix is kept for that reason: renaming would have broken 148 references between the paper, the supplementary appendices and the supplementary material.

## **Appendix B. Appendix B. Supporting tables**

These are the per-region and per-direction numbers the Results sections  
870 quote, given in full so that every claim can be checked against the values it

rests on rather than against a summary of them.

**Table B2. Signed univariate AUC of each predictor against burned, by region.** Primary natural-vegetation population; 10-cell ( $\sim 5$  km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never  
875 folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

| Feature      | Manavgat             | Bejís                | Muğla                | Evia                 | Montiferru           |
|--------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| elevation_   | <b>0.374</b> [0.289, | <b>0.643</b> [0.558, | <b>0.611</b> [0.532, | 0.541 [0.448,        | 0.584 [0.395,        |
| mean         | 0.471]               | 0.729]               | 0.690]               | 0.626]               | 0.762]               |
| slope_mean   | 0.531 [0.423,        | 0.521 [0.439,        | <b>0.637</b> [0.582, | 0.487 [0.418,        | <b>0.652</b> [0.506, |
|              | 0.642]               | 0.605]               | 0.686]               | 0.554]               | 0.771]               |
| ndvi_mean    | <b>0.636</b> [0.587, | 0.559 [0.497,        | <b>0.662</b> [0.616, | <b>0.639</b> [0.575, | 0.586 [0.450,        |
|              | 0.676]               | 0.619]               | 0.704]               | 0.701]               | 0.704]               |
| lst_anomaly_ | 0.482 [0.428,        | <b>0.418</b> [0.364, | 0.485 [0.395,        | <b>0.640</b> [0.567, | 0.395 [0.285,        |
| mean         | 0.530]               | 0.480]               | 0.566]               | 0.710]               | 0.535]               |
| current_lst_ | 0.538 [0.452,        | 0.477 [0.401,        | <b>0.325</b> [0.271, | <b>0.377</b> [0.301, | 0.370 [0.248,        |
| mean         | 0.621]               | 0.547]               | 0.382]               | 0.456]               | 0.513]               |
| current_     | 0.552 [0.460,        | 0.517 [0.429,        | <b>0.336</b> [0.275, | <b>0.362</b> [0.285, | <b>0.356</b> [0.233, |
| tvdi_mean    | 0.641]               | 0.595]               | 0.398]               | 0.442]               | 0.499]               |
| tvdi_        | 0.449 [0.391,        | 0.512 [0.443,        | 0.490 [0.396,        | 0.519 [0.444,        | <b>0.378</b> [0.282, |
| difference_  | 0.505]               | 0.583]               | 0.575]               | 0.589]               | 0.497]               |
| mean         |                      |                      |                      |                      |                      |
| downscaled_  | 0.552 [0.466,        | 0.484 [0.400,        | <b>0.307</b> [0.253, | <b>0.377</b> [0.297, | 0.365 [0.240,        |
| lst_mean     | 0.637]               | 0.560]               | 0.366]               | 0.459]               | 0.511]               |
| fused_lst_   | 0.540 [0.454,        | 0.481 [0.404,        | <b>0.325</b> [0.272, | <b>0.376</b> [0.300, | 0.370 [0.248,        |
| mean         | 0.622]               | 0.551]               | 0.383]               | 0.456]               | 0.513]               |

**What counts as a reversal.** A pair of regions is called a reversal  
880 only when their signed associations point to opposite sides of 0.5 **and each region's own interval excludes 0.5**. That is stricter than requiring the two regions' intervals to be disjoint, and the difference matters: for `current_lst_mean` between Manavgat and Muğla the two intervals are disjoint, but

Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no  
 885 established direction to reverse from. That pair is a point reversal, not a  
 supported one.

**Table B3. The cross-region reversals that meet the stricter criterion.** Difference intervals are from the same paired bootstrap.

| Feature                       | Region A | AUC   | Region B | AUC   | Difference [95 %<br>CI]    |
|-------------------------------|----------|-------|----------|-------|----------------------------|
| <code>elevation_mean</code>   | Manavgat | 0.374 | Bejís    | 0.643 | +0.269 [+0.138,<br>+0.397] |
| <code>elevation_mean</code>   | Manavgat | 0.374 | Muğla    | 0.611 | +0.235 [+0.102,<br>+0.360] |
| <code>lst_anomaly_mean</code> | Bejís    | 0.418 | Evia     | 0.640 | +0.221 [+0.123,<br>+0.313] |

890 Three pair-level reversals across **two** features, which is why Section 3.11  
 removes exactly those two. Twenty-nine further pairs reverse at the point  
 estimate only, spread across eight of the nine features, and they are not  
 counted. The conservative criterion costs the paper findings rather than  
 manufacturing them: a difference interval on the pair, which is the instrument  
 895 Appendix A(m) uses, would support more reversals than the three listed here.

#### *Appendix B.1. B4. The transfer matrix in precision-recall terms*

ROC-AUC is reported throughout the main text for comparability with  
 the susceptibility literature. A susceptibility surface is used as a ranked  
 area budget, so precision-recall is the operational quantity, and at target  
 900 prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same  
 frozen step9b exports as Table B9.

**Table B4. Thermal transfer, PR-AUC against the no-skill baseline.** The baseline is the target’s burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

| Direction             | ROC-AUC      | PR-AUC       | No-skill     | Lift        |
|-----------------------|--------------|--------------|--------------|-------------|
| Evia → Manavgat       | 0.686        | 0.094        | 0.038        | <b>2.45</b> |
| Bejís → Montiferru    | 0.594        | 0.289        | 0.212        | 1.36        |
| Evia → Montiferru     | 0.647        | 0.283        | 0.212        | 1.34        |
| Bejís → Muğla         | 0.619        | 0.093        | 0.070        | 1.33        |
| Muğla → Evia          | 0.653        | 0.379        | 0.287        | 1.32        |
| Montiferru → Bejís    | 0.548        | 0.093        | 0.072        | 1.28        |
| Montiferru → Muğla    | 0.619        | 0.089        | 0.070        | 1.27        |
| Evia → Muğla          | 0.577        | 0.086        | 0.070        | 1.23        |
| Muğla → Bejís         | 0.583        | 0.088        | 0.072        | 1.22        |
| Manavgat → Evia       | 0.613        | 0.343        | 0.287        | 1.20        |
| Manavgat → Montiferru | 0.533        | 0.243        | 0.212        | 1.15        |
| Montiferru → Manavgat | 0.567        | 0.043        | 0.038        | 1.11        |
| Montiferru → Evia     | 0.586        | 0.317        | 0.287        | 1.11        |
| Muğla → Montiferru    | 0.531        | 0.214        | 0.212        | 1.01        |
| Bejís → Manavgat      | 0.444        | 0.034        | 0.038        | <b>0.90</b> |
| Manavgat → Muğla      | 0.470        | 0.063        | 0.070        | <b>0.90</b> |
| Evia → Bejís          | 0.448        | 0.059        | 0.072        | <b>0.82</b> |
| Bejís → Evia          | 0.383        | 0.226        | 0.287        | <b>0.79</b> |
| Muğla → Manavgat      | 0.401        | 0.029        | 0.038        | <b>0.75</b> |
| Manavgat → Bejís      | 0.326        | 0.049        | 0.072        | <b>0.67</b> |
| <b>Mean</b>           | <b>0.541</b> | <b>0.156</b> | <b>0.136</b> | <b>1.16</b> |

Six directions fall below their own no-skill baseline, and only one exceeds

twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.4 shows  
910 that this count is largely a property of the evaluation frames rather than of a reversed predictor-burning relationship.

*Appendix B.2. B5. The same-geography event pair, in full*

Appendix A(m) reports this arm and Section 4.4 withdraws its **elevation** reversal as a frame artefact; Appendix A(o) explains why the thermal  
915 channels of this arm carry no verdict either way. The per-feature values are kept here because the arm is what motivated the frame test, and because its structural asymmetries have no analogue in the twenty-direction matrix.

**Table B5. Signed univariate feature-burned AUC, Muğla 2021 versus 2022.** Raw AUC against **burned**, never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ ; 10-cell ( $\approx 5$  km) spatial-block bootstrap, 1,000 replicates, seed 42  
920 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here  
925 fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

| Feature               | 2021 AUC [95 %<br>CI]           | 2022 AUC [95 %<br>CI]           | Reversal                        |
|-----------------------|---------------------------------|---------------------------------|---------------------------------|
| <b>elevation_mean</b> | <b>0.611 [0.532,<br/>0.690]</b> | <b>0.296 [0.230,<br/>0.355]</b> | <b>bootstrap-<br/>supported</b> |
| current_lst_mean      | 0.325 [0.271,<br>0.382]         | 0.515 [0.434,<br>0.580]         | point only                      |
| current_tvdi_mean     | 0.336 [0.275,<br>0.398]         | 0.594 [0.475,<br>0.674]         | point only                      |
| downscaled_lst_mean   | 0.307 [0.253,<br>0.366]         | 0.508 [0.435,<br>0.571]         | point only                      |
| fused_lst_mean        | 0.325 [0.272,<br>0.383]         | 0.519 [0.436,<br>0.583]         | point only                      |
| ndvi_mean             | 0.662 [0.616,<br>0.704]         | 0.707 [0.624,<br>0.777]         | none                            |
| slope_mean            | 0.637 [0.582,<br>0.686]         | 0.558 [0.468,<br>0.634]         | none                            |
| lst_anomaly_mean      | 0.485 [0.395,<br>0.566]         | 0.380 [0.249,<br>0.502]         | none                            |
| tvdi_difference_mean  | 0.490 [0.396,<br>0.575]         | 0.397 [0.265,<br>0.514]         | none                            |

*Appendix B.3. B6. The evaluation frames, and the signed associations they produce*

Section 4.4 states these results; the per-region values are here.

**Table B6. Evaluation-frame geometry of the five study regions.**

Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

| Region     | cells  | burned | median distance to<br>burned | share beyond 10 km |
|------------|--------|--------|------------------------------|--------------------|
| Manavgat   | 20,511 | 784    | 13.4 km                      | <b>60.1 %</b>      |
| Bejís      | 15,190 | 1,100  | 13.5 km                      | <b>63.1 %</b>      |
| Muğla      | 41,730 | 2,911  | 11.3 km                      | 55.3 %             |
| Evia       | 9,298  | 2,664  | 8.0 km                       | 43.7 %             |
| Montiferru | 2,544  | 539    | 2.7 km                       | <b>2.1 %</b>       |

**Table B7. Signed univariate AUC, frame as drawn against a 10**  
**km collar.** Point estimates; the intervals that decide the reversal question  
are given in the text below and in `collar_frame_bootstrap.csv` (10-cell  
blocks, 1000 replicates, seed 42). Signed and never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 is a direction, not weakness. The collar drops no  
burned cells in any region.

| Signed<br>univariate<br>AUC            | Manavgat     | Bejís | Muğla | Evia  | Montiferru | straddles 0.5 |
|----------------------------------------|--------------|-------|-------|-------|------------|---------------|
| elevation, full<br>frame (Table<br>B2) | <b>0.374</b> | 0.643 | 0.611 | 0.541 | 0.584      | <b>yes</b>    |
| elevation, 10<br>km collar             | 0.561        | 0.614 | 0.606 | 0.648 | 0.581      | no            |
| current LST,<br>full frame             | <b>0.538</b> | 0.477 | 0.325 | 0.377 | 0.370      | <b>yes</b>    |
| current LST,<br>10 km collar           | 0.386        | 0.405 | 0.332 | 0.286 | 0.376      | no            |
| current<br>TVDI, full<br>frame         | <b>0.552</b> | 0.517 | 0.336 | 0.362 | 0.356      | <b>yes</b>    |
| current<br>TVDI, 10 km<br>collar       | 0.392        | 0.454 | 0.342 | 0.250 | 0.361      | no            |

**Table B8. Region summary: populations and gate outcomes.** Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled pop-  
950 ulation, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

| Region                     | Total cells | Valid cells | Burned | Prevalence (all valid) | TSG cells | Burned in TSG | TSG prevalence | Burned natural-veg fraction | Gate verdict |
|----------------------------|-------------|-------------|--------|------------------------|-----------|---------------|----------------|-----------------------------|--------------|
| Manavgat 2021              | 24,150      | 24,087      | 796    | 0.033                  | 20,511    | 784           | 0.038          | 0.984                       | pass         |
| Bejís 2022                 | 15,759      | 15,759      | 1,103  | 0.070                  | 15,190    | 1,100         | 0.072          | 0.991                       | pass         |
| Muğla 2021                 | 73,098      | 73,045      | 3,026  | 0.041                  | 41,730    | 2,911         | 0.070          | 0.958                       | pass         |
| North Evia 2021 (extended) | 22,925      | 22,906      | 2,788  | 0.122                  | 9,298     | 2,664         | 0.287          | 0.945                       | pass         |
| Montiferru 2021            | 3,234       | 3,173       | 697    | 0.220                  | 2,544     | 539           | 0.212          | 0.723                       | pass         |

**Table B9. Cross-region transfer matrix, thermal model, TSG population.** Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ( $\lambda = 10^{-5}$ ).

| Direction           | Raw                  | Region-wise z-score  | CORAL                |
|---------------------|----------------------|----------------------|----------------------|
| Manavgat→Bejís      | 0.326 [0.305, 0.349] | 0.477 [0.451, 0.502] | 0.511 [0.484, 0.534] |
| Bejís→Manavgat      | 0.444 [0.408, 0.480] | 0.457 [0.420, 0.497] | 0.555 [0.528, 0.583] |
| Manavgat→Muğla      | 0.470 [0.451, 0.488] | 0.431 [0.411, 0.449] | 0.443 [0.423, 0.461] |
| Muğla→Manavgat      | 0.401 [0.378, 0.426] | 0.559 [0.531, 0.587] | 0.560 [0.535, 0.587] |
| Manavgat→Evia       | 0.613 [0.593, 0.631] | 0.542 [0.520, 0.565] | 0.539 [0.518, 0.561] |
| Evia→Manavgat       | 0.686 [0.653, 0.716] | 0.516 [0.489, 0.544] | 0.527 [0.500, 0.553] |
| Bejís→Muğla         | 0.618 [0.601, 0.635] | 0.518 [0.501, 0.535] | 0.507 [0.489, 0.524] |
| Muğla→Bejís         | 0.583 [0.561, 0.607] | 0.535 [0.512, 0.557] | 0.560 [0.538, 0.581] |
| Bejís→Evia          | 0.383 [0.363, 0.402] | 0.532 [0.509, 0.551] | 0.499 [0.479, 0.518] |
| Evia→Bejís          | 0.448 [0.426, 0.470] | 0.549 [0.524, 0.575] | 0.549 [0.525, 0.573] |
| Muğla→Evia          | 0.653 [0.636, 0.671] | 0.561 [0.543, 0.580] | 0.563 [0.545, 0.582] |
| Evia→Muğla          | 0.577 [0.560, 0.593] | 0.501 [0.485, 0.518] | 0.530 [0.515, 0.546] |
| Montiferru→Manavgat | 0.567 [0.539, 0.594] | 0.573 [0.538, 0.609] | 0.606 [0.574, 0.639] |
| Manavgat→Montiferru | 0.533 [0.488, 0.580] | 0.586 [0.540, 0.629] | 0.592 [0.550, 0.631] |
| Montiferru→Bejís    | 0.548 [0.521, 0.578] | 0.574 [0.552, 0.596] | 0.569 [0.548, 0.591] |
| Bejís→Montiferru    | 0.594 [0.560, 0.631] | 0.550 [0.500, 0.601] | 0.574 [0.530, 0.621] |
| Montiferru→Muğla    | 0.619 [0.604, 0.634] | 0.576 [0.562, 0.589] | 0.565 [0.550, 0.579] |
| Muğla→Montiferru    | 0.531 [0.495, 0.568] | 0.587 [0.549, 0.624] | 0.584 [0.547, 0.623] |
| Montiferru→Evia     | 0.586 [0.565, 0.606] | 0.630 [0.611, 0.649] | 0.624 [0.605, 0.641] |
| Evia→Montiferru     | 0.647 [0.608, 0.682] | 0.568 [0.528, 0.609] | 0.581 [0.539, 0.623] |

## Appendix C. Appendix C. Limitations

Section 5.7 states the four limitations that bind the conclusions; all ten  
960 are here. The rest of this appendix, which specifies the protocol, is released  
with the supplementary appendices rather than printed here.

### *Appendix C.1. C.5 Limitations, in full*

Section 5.7 states the four that bind the conclusions; all ten are here.

(i) **No meteorological covariates** enter the models, so we cannot say  
965 how local skill and portability behave for a mixed thermal-plus-weather pre-  
dictor set.

(ii) **The same-geography comparison covers one region only**, and  
even there year and seasonal phase are confounded. That confound cannot  
be resolved in this study area, and the reason is specific: the two events sit  
970 42 days apart in median burn day-of-year, neither year contains a second  
event at the other’s phase, and a calendar-matched arm would carry nine  
burned cells against this design’s gate minimum of thirty. The positive-block  
count for the 2022 arm is separately below the floor this design sets itself,  
at eleven against sixteen (Appendix A(m)). Its 331 burned cells also leave  
975 the thermal reversals unresolved at interval level, and the pair holds place  
fixed but not population. Those two arms are additionally the only transfer  
directions here computed by us rather than read from the pipeline author’s  
frozen export, with his unmodified code and the same pinned environment.

(iii) **All labels derive from a single burned-area product**,  
980 MCD64A1 (Giglio et al., 2018), whose omission and commission character-  
istics (Boschetti et al., 2019) bound every model evaluated here. No second

burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

985 (iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires  
990 multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly  $\pm 0.02$  to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

995 (vii) **Manavgat’s atypical transfer behaviour remains unexplained.** It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed  
1000 association by more than +0.0003 (Appendix A(v), Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.4). With one fire season per region the remaining candidates are not separable in this design.

(viii) **The interval-support counts are less stable than the point**  
1005 **estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of the paired

thermal-minus-baseline deltas, ten positive, seven negative and three uncertain, turns on a lower bound of  $-0.00045$ . The point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that  
1010 leans on an exact count of supported directions should be read at that precision. (ix) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). We report the equalised arm alongside the frame-as-drawn arm rather than replacing one with the other, because the collar radius is itself a choice and 5 km and 10 km do not  
1015 agree exactly (0.608 against 0.616). The deeper limitation is that the frames were fixed upstream of this work, in **repo/**, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Nor can their independence from outcomes be fully documented: only Montiferru’s box is derived by a rule,  
1020 and for Manavgat, Bejís and Muğla the version history cannot show the box fixed before the first gate result (Section 3.1). Any future cohort should fix the frame by an explicit accessible-area rule (Barve et al., 2011) before any predictor is computed, and we treat that as the main design lesson of this paper.

(x) **Other classifiers are compared by point estimate only.** The  
1025 headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Appendix A(h) shows the transfer result is not an artefact of that choice: three further estimators, including a penalised linear one, all land between  
1030 0.510 and 0.556 and all place fourteen of twenty directions above chance. Those are point estimates without intervals, so the ordering among them is

not claimed as a result. Estimator classes beyond these four were not tried, and a different inductive bias might behave differently, but across the four tried the negative result is a property of the predictors rather than of an  
1035 unregularised estimator.

## Software and data availability

**Analysis code and frozen outputs.** Name: `thermal-twin`. Developers: Y. E. Cogurcu and E. Metin; contact: `ycogurcu@cu.edu.tr`. First available: 2026. Hardware: a standard desktop computer. Software  
1040 required: Python 3.12 with NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, the versions every reported number was produced with; scikit-learn in particular carries a cross-region tolerance of about  $\pm 0.02$  to 0.03 between versions (Appendix C.5(vi)). Program language: Python. Size: about 21 MB. Availability: <https://github.com/dryuemco/thermal-twin>,  
1045 which holds the scripts that regenerate each reported artefact, the frozen numeric outputs behind every table and figure, and the supplementary appendices (Appendix A and protocol sections C.1 to C.4, C.6 and C.7, at [paper/supplementary\\_appendices.md](#)). Cost: free.

**Upstream processing pipeline.** Name: `satellite-thermal-digital-twin`. Developer: E. Metin. First available: 2026. Hardware: a standard desktop computer; the satellite exports run on Google Earth Engine and need an Earth Engine account. Program language: Python. Size: about 20 MB. Availability: <https://github.com/emrehann17/satellite-thermal-digital-twin>, MIT  
1050 licence; the commit of record for every number reported here is `6381f4c`.  
1055

Three components once outside the release are now inside it: the reproduction-check driver and its validation package, on which Section 3.13 rests; the few-shot run, reachable through the tag `few-shot-run-19d825b`; and the ERA5-Land diagnostic, whose manifest records commit `a07ea33`.

1060 Cost: free.

**Data.** All satellite inputs are public and were retrieved through Google Earth Engine: burned-area labels from MODIS MCD64A1 Collection 6.1, land cover from ESA WorldCover, and the thermal, optical and terrain inputs described in Section 3.4; no proprietary or restricted data were used. The  
1065 modelling dataset each number rests on is identified in the pipeline by SHA-256, which is how the one provenance incident in this project was settled: a rebuild had replaced one region’s file at its canonical path, and the recorded hash identifies the frozen original unambiguously. No digital object identifier is minted and no archival deposit exists.

## 1070 **Declarations**

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**Competing interests.** The authors declare no competing interests.

**Ethics approval.** Not applicable. This study involved no human participants, animal subjects, or personally identifiable data.

**Author contributions.** Stated in CRediT terms. **Emrehan Metin:**  
1085 Software, Data curation, Investigation, Validation, Writing – review and editing. **Yunus Emre Cogurcu:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision.

## 1090 **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work the authors used Claude (Anthropic) in order to write and run analysis and verification code against the frozen pipeline outputs, cross-check reported numbers against those outputs, and draft and edit manuscript text. After using this tool, the authors reviewed  
1095 and edited the content as needed and take full responsibility for the content of the publication.

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